

faster than our eyes can register. The fake approval happens before we can even see the prompt.

T3: Bluetooth HID fallback. If access fails via USB, attackers may switch to Bluetooth, hijacking paired accessories—such as headsets—to inject commands wirelessly.

Why this should scare you

This is not science fiction. It is real, tested, and terrifyingly simple. Current USB security assumes the user is in control. Choicejacking breaks that assumption completely. More worryingly, antivirus tools are unable to detect it. These commands happen below the software layer—our phone treats them as normal hardware input.

If You're an Engineer, Read This Twice. If we design phones, infotainment systems, USB accessories, or embedded panels, fundamental changes are needed. To defend against attacks like choicejacking:

Do not trust HID inputs by default. It is no longer enough to verify what a device is saying; we must verify who is saying it. This means implementing identity checks or secure pairing for connected peripherals. Next, permission prompts must be redesigned to withstand spoofing.

Simple pop-ups won't suffice; use biometric confirmation or secure touchscreen zones that are protected from fake taps. High-risk actions, like file access or enabling developer modes, should never execute instantly. Introducing even a brief delay can help block timing-based attacks that exploit microsecond windows. Finally, block all USB data access during boot-up; until a device is fully unlocked, it should accept

power only, not data, preventing rogue inputs from executing before the system is ready. And if you are working on silicon-level defences, now is the time for a zero-trust USB model, no access without verified cryptographic identity.

For users: Smart habits, not just smart devices

Until hardware catches up, we can protect ourselves by following simple habits:

- Never use public USB ports—they are untrusted computers
- Carry our own charger or power bank
- Use a USB data blocker (a device that blocks data pins and allows power only)
- Select 'Charge Only' every time we plug in, especially on Android
- Enable lockdown mode on iOS or Android
- Keep our OS updated

Where the industry stands (and falls short)

Choicejacking is not just a user threat—it is a design challenge for the electronics ecosystem. Systems can no longer trust what a device claims to be; they must verify who is sending the command. USB and Bluetooth permission flows need secure, spoof-proof visual cues. Biometric confirmation, time delays, and hardened permission pathways can slow down or block automated attacks. The current trust model—where any plugged-in HID is assumed legitimate—is far too vulnerable.

And this is likely only the beginning. As devices become more connected and ports more multi-functional, attackers will find new

ways in. Rumours already suggest malicious USB-C cables capable of launching similar attacks while appearing completely normal. Without stronger

TABLE 1
ENGINEER'S CHECKLIST

Area	Suggested Action
USB stack	Add HID source validation
Firmware	Disable HID until trust is verified
UI design	Add time delays and biometric confirmations
Hardware	Use USB filters/firewalls with policy enforcement
Product defaults	Data off by default; user must opt in

security, the convenience we rely on today could become tomorrow's vulnerability.

There is some good news. Android has introduced USB restrictions, though input validation remains weak. Apple's Lockdown Mode offers partial protection but does not address HID spoofing. Linux tools such as USBGuard show promise but operate only at the software level, leaving hardware exploits untouched. What is really needed is a Trusted HID protocol: a secure, cryptographic handshake between device and host, similar to Bluetooth LE's encrypted pairing.

Until such standards are adopted, it is up to all of us—designers, OEMs, and users—to stay ahead of the threat. When the attack resembles a charger, the only way to stay safe is to reevaluate trust at every level, including chips, OS, and habits.

The age of blind USB trust is over. Choicejacking does not exploit a bug; it exploits a design assumption. The way forward is a fundamental rethink of how we handle USB input. This is our wake-up call. Whether we are engineers, developers, or simply people charging a phone in an airport lounge, we must not trust the port unless we own it. Smarter systems. Smarter habits. No free passes. In today's world, the most dangerous cable may be the one that claims it is just charging. **EFY**

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TABLE 2
KEY TAKEAWAYS

What	How it works	What to do
Choicejacking	Spoofs inputs to bypass USB prompts	Avoid public ports
Attack time	Under 133ms	Use data blockers
Affects	Android and iOS devices	Update your OS
Real threat?	Not yet in wild, but proven	Be proactive